Ranking and Selection with Probing

1 Problem Formulation

The optimization problem we wish to solve is

$$\max_{z\in\{0,1\}^m} \left\{ -\alpha^T z + \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i,\xi) \mid \{\eta_j \colon z_j = 1\} \right] \right] \right\}. \tag{1}$$

- There is a set of k alternatives to choose from, denoted by $\mathcal{X} = \{x_1, x_2, \ldots, x_k\}$. The notation x_i is not strictly necessary (i.e., neither the value or dimension of x_i matter), but we can keep it for now.
- $\eta = \{\eta_1, \eta_2, \ldots, \eta_m\}$ represent a (small) set of variables that can be probed, with corresponding costs $\alpha_1, \alpha_2, \ldots, \alpha_m$. Other probing work [3] assumes that the probing cost function $\alpha(z)$ is monotone-increasing and (in one setting) (sub?)modular and use this property in their experiments. This generalization beyond additive cost functions may not be important in our case.
- ξ represents the set of all random primitives needed to perform one replication of the simulation model, which outputs a scalar $Y(x_i, \xi)$ when simulating alternative x_i. In simulation, ξ is typically some abstract object and generally not well-defined. For example, it could be the random variates generated within a replication (e.g., random demands), the Uniform(0, 1) random variates required to generate these random variates, or even the *seed* of the random number generator used on that replication. The dimension of ξ may be very large and random (e.g., the number of people who call into a call center on a given day), so it is generally hard to work directly with ξ in the simulation context. In some of the work on probing, it is convenient for their to be a one-to-one correspondence between components of η and components of ξ, implying they have the same dimension. In other cases, such as some experiments in [3], probing an η_j reveals a corresponding ξ_j. This setup does not make much sense in the simulation context. [bp: To check my understanding: at the end of the day, you just need the capability to generate n_i simulations for each of the k alternatives, so you have a bunch of Y_{ij} that you can (at least!) compute $\hat{\mu}_i$ and $\hat{\sigma}_i$]

  [de: What we require is the ability to generate i.i.d. replications of $Y(x_i, \xi)$ conditional on $\{\eta_j \colon z_j = 1\}$. This often means being able to generate ξ

conditional on $\{\eta_j \colon z_j = 1\}$ and then input this ξ into the simulation model and evaluate the output.]

[bp: I really liked your comment that this setup does not make much sense in the simulation setting. In Stochastic Programming, Y is the uncertainty, and it is written down as part of the model, in equations such as inventory dynamics.]

Observations about Problem (1):

- The inner problem

$$\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\}\right]$$

  is a classical ranking-and-selection problem (R&S), but parameterized by the realization of the probed η_js, which we can think of as a *context* or *covariates*. In the simulation sphere, this problem cannot be solved exactly, but many (sample-)efficient algorithms exist that take simulation replications until they can guarantee to select the optimal x (or a δ-optimal x) with high probability, called a probability of correct selection (PCS) (respectively, probability of good selection (PGS)) guarantee. Others work with a fixed budget of simulation replications and seek to maximize PCS. Either way, we should think of the inner problem as well-studied and solvable, assuming k is not too large and the per-replication run time is not too large. There also exist a small number of R&S algorithms that seek to minimize the expected optimality gap of the selected x; see [1], for example. These algorithms are less in vogue, but would seem to be more relevant to us than PCS/PGS ones, especially considering how optimality gaps naturally arise in this context (see upcoming discussion).

- Special cases include the R&S problem associated with no probing (i.e., the original R&S problem)

$$\max_{i=1,2,\ldots,k} \mathbb{E}[Y(x_i, \xi)] \tag{2}$$

  and full probing (i.e., perfect information)

$$\max_{i=1,2,\ldots,k} \mathbb{E}[Y(x_i, \xi) \mid \eta_1, \eta_2, \ldots, \eta_m]. \tag{3}$$

- The problem formulation requires the ability to sample from the distribution of $\xi \mid \eta$ for all subsets of η. Given the challenges or working directly with ξ, rather than attempt to explicitly model the joint distribution of ξ and η, a proposal would be to assume something like

$$\xi \mid \eta \sim G(\theta(\eta)) \tag{4}$$

  where G is a stochastic generator with parameters $\theta(\eta)$. The term *stochastic generator* is non-standard, but is an attempt to simultaneously consider probability distributions and stochastic processes. As for concrete

examples, we might have that [bp: Question to Dave: When you think about this black-box simulator, what is complicated about it? It seems that the distributions can be standard, but the relationship between an alternative i and the black box's output is just IMPOSSIBLY nonlinear, right? There is no model written down in equations that connects Y to x and ξ, that is the whole purpose of the broad $Y(x, \xi)$ notation, I believe.] [de: That's right. The random primitives (the ξs) can be simple, but the mapping $Y(x, \xi)$ is assumed to be highly non-linear and defies analytical approaches.]

  - given a realization of market performance, demand for a product is normally distributed according to $\xi \sim \text{Normal}(\mu(\eta), \sigma^2(\eta))$, or
  - given a forecast for tomorrow's temperature, the arrival process of customers to an ice cream stand is $\xi \sim \text{PoissonProcess}(\lambda(\eta))$, where $\lambda(\eta) = \lambda_0 + \beta^T \eta$.

  The first of these examples shows how observing η might make ξ less variable (by reducing the variance of the normal distribution) and, by extension, $Y(x_i, \xi)$ less variable for certain x_i. The parameterization in (4) may also apply in the financial engineering context where the randomness ξ follows a Geometric Brownian motion with certain drift and diffusion parameters. In other, very special (low dimensional) cases, it may be possible to explicitly model the joint distribution of ξ and η. That approach would be harder to justify in the simulation context.

  In the model in (4), ξ unconditionally follows a distribution with random parameters, i.e., a *compound* distribution. These can arise, for example, in doubly stochastic arrival processes (like in the second example). It can be shown (via the Law of Total Variance) that the additional randomness in the parameters leads to additional variability in ξ.

- The (linear) correlation between ξ and η is not as important as the effect that the inputs η have on the output Y and how they change the optimal value of the resulting R&S problem when their conditional distribution changes (conditional on components of η). In other words, the probing variables that make ξ (or components of ξ) less variable may not translate to improvements in the objective function value, i.e., exploitable information about how to pick x given η.

- Something overlooked so far in this discussion is that with this notation, the uncertainty ξ depends on the probing decision η (and/or z), but does not (evidently) depend on the second stage decision x. A strength of simulation optimization (like R&S) is that ξ can depend on x - as long as we can simulation $\xi \mid x$, the R&S algorithm will take care of the rest. Here, there may not be any loss to including x in the parameterization of the distribution of ξ. It's easier to start by assuming ξ does not depend on x, but we might be able to allow this dependence if we end up using existing

R&S technology to solve some sub-problem. [bp: This observation made me write a full appendix for these notes! Please see Appendix A.]

[de: If we wish to handle decision-dependent uncertainty, we would need to be able to generate ξ conditional on both $\{\eta_j \colon z_j = 1\}$ and x_i, which means we have different conditional distributions for each x_i.]

- Problem (1) has the same optimizer as the problem

$$\max_{z \in \{0,1\}^m} \left\{ -\alpha^T z + \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right] - \mathbb{E}[Y(x^*, \xi)] \right\}. \tag{5}$$

  where x^* denotes the optimal solution to the R&S problem with no side information, (2). To see this, notice that we have subtracted the constant $\mathbb{E}[Y(x^*, \xi)]$, which does not depend on z. Furthermore, by the Tower Rule,

$$\mathbb{E}[Y(x^*, \xi)] = \max_{i=1,2,\ldots,k} \mathbb{E}[Y(x_i, \xi)] = \max_{i=1,2,\ldots,k} \mathbb{E}_\eta[\mathbb{E}[Y(x, \xi) \mid \{\eta_j \colon z_j = 1\}]]$$

  for any $z \in \{0,1\}^m$. Thus, putting aside the costs of the probing variables, we are interested in those components of η that lead to large differences

$$\mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right] - \max_{i=1,2,\ldots,k} \mathbb{E}_\eta[\mathbb{E}[Y(x, \xi) \mid \{\eta_j \colon z_j = 1\}]]. \tag{6}$$

  In other words, which probing variables give us the greatest benefits from being able to defer our selection decision until *after* we have seen them? (Notice the reversed orderings of the expectation and the maximization.) In stochastic programming, (6) is called the Expected Value of Perfect Information (EVPI) and captures the benefit of making a decision in the second stage rather than the first stage. It is easy to show that EVPI ≥ 0 for all $z \in \{0,1\}^m$ (Birge & Louveaux, Proposition 1, page 140). This means there is never a downside to having more side information. Side information gives us the opportunity to better tailor our selection to the problem instance, i.e., to the realization of η. Note: EVPI is more commonly used as a post hoc metric. Computing EVPI typically relies on there being a finite number of scenarios or some other special problem structure (e.g., newsvendor).

- Similarly, we should be able to show that the objective function in (1) is *submodular* using the linearity of the cost function and properties of conditional expectation. Other than giving theoretical bounds on the performance of a greedy algorithm, would showing that (1) is submodular offer any benefits in terms of designing solution methods?

- Consider a greedy algorithm that sequentially chooses to reveal probing variables until all have been probed. The reward function for the greedy algorithm might be to next probe the variable that allows us to gain the most in objective value by being able to pick an alternative other than x^*

after subtracting the probing cost. This involves looking at the *optimality gap* between x^* and $x^*(\eta)$, where $x^*(\eta)$ is the solution to the problem parameterized by η. For the first step of the greedy algorithm, the reward function might look like

$$r(j) = -\alpha_j + \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j\}\right] \right] - \max_{i=1,2,\ldots,k} \mathbb{E}[Y(x, \xi)]$$

for all $j = 1, 2, \ldots, m$, and for subsequent stages it could look like

$$r(j) = -\alpha_j + \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{H \cup \eta_j\}\right] \right] - \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid H\right] \right]$$

for all $j \notin H$, where H is the set of variables that have already been probed. The greedy algorithm would sequentially add the unprobed variable j with the maximum reward, $\arg\max_{j \notin H} r(j)$, until the rewards of the unprobed variables are all negative or all variables have been probed. Greedy algorithms can be very expensive in the stochastic programming domain, so we shouldn't take it for granted that a greedy algorithm (or approximation thereof) is implementable.

- There might be a way to exploit prior work on R&S with covariates/contexts, but the probing R&S problem has important differences. For example, our objective factors into how much of an *optimality gap* we can create by being able to make our selection of x after observing some components of η. Each realization of the probed variables can be thought of as a context, which has an associated R&S problem [bp: I agree with that. More and more I feel that contextual that is not free = probing]. The optimize-then-weight (OtW) approach of [2] may struggle to quantify this optimality gap, though some of the (yet unpublished) follow-up work concerns high probability bounds for the optimality gap associated with some predicted optimal x. The weight-then-optimize (WtO) approach could make it easier to estimate optimality gaps, because one has estimates of $\mathbb{E}[Y(x_i, \xi) \mid H]$ for all alternatives i and all contexts.

- One could imagine running ranking and contextual selection (R&CS) to first solve a small number of R&S problems corresponding to different *designed* realizations of the full vector η. One could then come up with a 1-NN predictor for the optimal alternative for any realization of a subset of the probing variables. (For example, if we are considering a problem with a single probing variable, the full contexts are projected down to this one-dimensional space and a new 1-NN model is constructed on the projected space.) To estimate the optimality gap (and overall objective from probing that variable), one needs to weight the problem instances according to the probabilities that the probed variable takes those values. When considering different combinations of probed variables, we apply different projections and weightings, depending on which variables are probed. [bp: Yeah, this makes a lot of sense. I see it as using [2] and doing a sort of

brute force, exhaustive method to cope with all probing possibilities. Not elegant, but a decent starting point/benchmark.] [de: I had assumed that we could pre-solve only problem instances with perfect information, but I no longer think the projection idea will work. If not, then it would become more like a brute-force approach, testing realizations from the 2^m different probing combinations, which seems far too expensive.] The problem with this approach is that estimating optimality gaps is not straightforward if applying the OtW approach of [2].

2 Questions about the Motivating Context

- Are decisions to be made repeatedly over time? And do we pay once up front for the probing variables? Or are we paying for side information for just a one-time R&S problem? The simplest setting is making repeated decisions over time, where we pay for *the ability to probe* components of η once up front and routinely observe the realizations of the probed variables each time we need to make a decision. In other words, a one-time payment, but benefits over time. The setting of repeated-decisions-over-time aligns with the expected value over η in (1) because we incur different rewards (reward = benefit over always picking x^*) each time we make a decision. If we imagine making T decisions over time, we would need to include it in the objective function so that the costs α are weighed against the cumulative rewards. (This might just mean setting $\alpha_j \leftarrow \alpha_j/T$.) There are some motivating applications where probing is done once but repeated decisions are made based on the latest information. For example, the decision-maker may pay for a subscription, or a drilling machine, or a sensor that yields probed information (realizations of η) over time and decisions are made on a recurring basis.

- Can we say that the problem with expected value plus probing represents sort of a risk-aversion, without having a risk measure in place? In that sense, is it a more appropriate framework when dealing with one-off decisions, e.g, exploration of a mine, construction of a tunnel, etc?

- [bp: Bernardo's trivial comment] I read [7], and as it seems to be a tradition in the R&S community, the results are derived under normality assumption. One nice thing about probing is that if we assume the η vector is normal as well, then we have normality in the conditional distribution:

  Let
  $$\xi \in \mathbb{R}^p, \qquad \eta \in \mathbb{R}^q,$$
  and assume that
  $$\begin{bmatrix}\xi \\ \eta\end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\mu_\xi \\ \mu_\eta\end{bmatrix}, \begin{bmatrix}\Sigma_{\xi\xi} & \Sigma_{\xi\eta} \\ \Sigma_{\eta\xi} & \Sigma_{\eta\eta}\end{bmatrix}\right),$$
  where $\mu_\xi \in \mathbb{R}^p$, $\mu_\eta \in \mathbb{R}^q$, and the block covariance matrix is partitioned accordingly.

Then the conditional distribution of ξ given η is multivariate normal:

$$\xi \,|\, \eta \;\sim\; \mathcal{N}\big(\mu_\xi + \Sigma_{\xi\eta}\Sigma_{\eta\eta}^{-1}(\eta - \mu_\eta),\;\; \Sigma_{\xi\xi} - \Sigma_{\xi\eta}\Sigma_{\eta\eta}^{-1}\Sigma_{\eta\xi}\big)\,.$$

[de: Its usually the outputs, the $Y(x,\xi)$s, that are assumed to be normally distributed in R&S, rather than the inputs ξ. Still, for financial engineering applications, joint normal distributions like this could arise when working with geometric Brownian motions. For example, η could be the price process of another asset. I'm struggling to think of why this information would not be public, unless we were thinking of something like a Bloomberg terminal that is given real-time information. The closer we get to real-time decision making, though, the harder it is to defend R&S as a workable solution method.] [bp: Good points. I have been working with contextual information in finance, and yeah, things are sort of free/available. I mean, you pay for some things here and there, but is it relevant in a portfolio of 10 million dollars? No. In the discussion with Franco we feel that we should look into more "physical" settings, in which probing is paying for expensive machines, doing field experiments, or even paying for satellite images. As of now, I think for finance free probing, aka contextual, seems to be more appropriate. See if you have time]

3 Connections

The problem appears to have connections to many related simulation methodologies, including variations of R&S.

- Problem (1) involves an expectation over realizations of the probed variables. This weakly resembles other R&S setups where there are multiple problem instances to integrate over, either under a Bayesian paradigm, e.g., the posterior integrated PCS (IPCS) of [5], or R&S with covariates (see next point).
- Connections to ranking and contextual selection: [2].
- Other connections to Bayesian R&S. Here one could think of the unconditional distribution of ξ as a prior distribution and $\xi \mid \eta$ as a posterior distribution.
- There may be connections to R&S with control variates [4], but there the control variates appear during the simulation run (in each replication) and are not correlated with the inputs of the simulation, etc. Control variates might be relevant here because we might think about our problem as trying to identify which probing variables are most strongly correlated with ξ. [bp: Uhhhh, I like that! When I think of control variates, I think of option pricing, which is a standard and successful domain in which control variates are applied. Of course, in that setting, the decision maker comes up with the control variate]

[bp: Follow-up: are there problems for which finding the control is hard? Or are there candidate control variables to be considered? Why would I pay for that?]

- Some connections to input uncertainty (IU) where the interest is on quantifying the uncertainty in outputs ($Y(x, \xi)$) due to uncertainty about the input model/distribution. Our case is different, because we are not so much trying to reduce uncertainty in the outputs by reducing uncertainty in the inputs, as much as we are trying to gain an advantage from having been able to optimize after seeing a realization of η.
- Connections to expected opportunity cost (EOC), which is essentially the expected optimality gap between the (random) alternative selected by a frequentist R&S procedure and the optimal alternative: [1].
- Connections to the (very new) two-stage R&S problem: [7]. From a quick skim of the paper, the first stage decision is not one of probing per se, though it still impacts the second stage. The methods proposed are fairly derivative of common R&S approaches: EVI and OCBA, so there it still an opportunity to propose something novel in this space. [de: I think this paper is the most important for us to read for now.] [bp: Reading it now!] [bp: Yeah, it does make a lot of sense. In our case, probing decision would be first-stage, and the x would be the second stage. I believe what they propose can be readily applied to problem (1)]
- [de: Is there perhaps a connection between a relaxation of this problem (where z can be real-valued) and functional principal components analysis (PCA)? The optimal z may mix probing variables in a way that seeks to maximize the variance/spread of the optimality gap.]

4 Next Steps

1. Come up with one or two concrete applications where this problem formulation makes sense. That way we can understand the dimension of the variables, the time available to make decisions, and how η and ξ are jointly related. The repeated-decision context with a subscription/sensor setup is a promising starting point.
2. Look more closely at [7] to see how it can be applied to our problem or serve as a benchmark.
3. Think about ways to *estimate* the optimality gap under different realizations of η. One (expensive) idea, is to build a predictive model of $\mathbb{E}[Y(x_i, \xi) \mid \eta]$ as a function of η for each x_i and then manipulate these estimates (e.g., weight them against the probabilities of different realizations of η). (Note: the dimension of the probed η can change, so there would likely be multiple models defined over nested spaces with differing dimensions.)

5 Likelihood-Ratio Reweighting

A first idea for approaching problem (1) is to not view it as a R&S problem. Sure, there is nominally an inner R&S problem, but existing R&S approaches for two-stage problems create many synthetic alternatives by crossing first-stage decisions with environments (scenarios) with second-stage decisions, which is computationally expensive; see, e.g., [7]. In our context, z would be the first-stage decision, η the environment, and x_i the second-stage decision, with ξ being the second-stage randomness that appears in [7].

An alternative approach is to obtain estimates of

$$\mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\}\right] \right] \text{ for all } z \in \{0,1\}^m$$

and then optimize over these estimates. For this approach to be tractable, we require a few things:

1. k is small enough that we can simulate all possible second-stage decisions (all x_is). An acceptable value of k will depend on the per-replication simulation cost, but let's pretend it's on the order of thousands, which would nowadays be considered a modest-sized R&S problem, given access to parallel computing, which this approach can likewise exploit.

2. We have access to the joint distribution of any combination of the η_js (as dictated by the z_js).

3. We have access to the conditional likelihood of ξ given $\{\eta_j \colon z_j = 1\}$ for any ξ and any realizations of η_js and any combinations of the z_js.

There are also a few notable things we don't require:

1. m does not need to be small (say, < 10), but this will certainly help with the subsequent optimization.

2. ξ does not need to have finite support. Nor does its dimension matter, other than its potential influence on the expression for the likelihood ratio.

3. We don't strictly need the doubly stochastic framework where $\xi \mid \eta \sim G(\theta(\eta))$, though in that setting, deriving the likelihood ratios will be relatively straightforward. In the doubly stochastic framework, we explicitly model the conditional distribution of ξ given η, whereas it should still be possible to apply the approach sketched out here if we instead explicitly model the joint distribution of ξ and η and only implicitly obtain the conditional likelihood. [de: There is a bigger discussion we should have about the user's modeling of the conditional and joint distributions. It's easy to imagine a user explicitly specifying a marginal distribution for η and a conditional distribution of ξ given η, and thereby implicitly fix a joint distribution for ξ and η. However, if there are different η (different probing variables), the resulting marginal distributions of ξ might not agree,

which is a problem. The way around this is probably to have the user specify the marginal distribution of η (as a vector) and the conditional distribution of ξ given the full η vector, and then work out the other conditional distributions (conditional on subvectors of η) by integrating out the unobserved components of η.]

4. We don't require decision-independent uncertainty. [de: The notation still reflects DIU, but I have added another version of (7) that shows the DDU case.] [bp: That is nice, and it could be a strong selling point. In the Stochastic Programming community, DDU is a hot topic these days, and the corresponding models are way harder than the base model. That does not seem to be the case here.]

Having said all that, the idea is fairly straightforward:

- Generate n i.i.d. realizations of ξ from the unconditional distribution (i.e., no probing). [de: This could be relaxed to some generic distribution that satisfies the assumptions of the LR method, namely, being absolutely continuous with respect to the conditional distributions of ξ given η, i.e., having a support that is a superset of the support of the conditional distributions. The advantages of using the marginal distribution of ξ is that it (by definition) satisfies this requirement and this distribution might already be explicitly modeled by the user.]

- Use the simulation model to evaluate $Y(x_i, \xi_\ell)$ for all $i = 1, 2, \ldots, k$ and all $\ell = 1, 2, \ldots, n$. This amounts to $k \times n$ simulation replications, which can be performed in parallel.

- Use likelihood-ratio reweighting to get unbiased estimates of

$$\mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\}\right]$$

  for any and all $\{\eta_j \colon z_j = 1\}$.

- Integrate the maximum (over $i = 1, 2, \ldots, k$) of these estimates over all $\{\eta_j \colon z_j = 1\}$, weighted by the likelihood of $\{\eta_j \colon z_j = 1\}$. The result will be a biased estimate of

$$\mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\}\right] \right]$$

  but we will hope that the bias is not severe. (The bias should vanish as $n \to \infty$, assuming the variance of the LR estimator is not infinity.)

- Optimize these biased estimates over all $z \in \{0, 1\}^m$ to find the optimal probing decision. Note: if the bias in the estimates is all in the same direction (and of similar magnitude), it should not be too harmful to the optimization.

Mathematically, the sample-average optimization problem we propose to solve is

$$\max_{z\in\{0,1\}^m} \left\{ -\alpha^\top z + \int_{\mathbb{R}^{d_z}} \left[\max_{i=1,2,\ldots,k} \frac{1}{n} \sum_{\ell=1}^{n} Y(x_i, \xi_\ell) \frac{f(\xi_\ell; \eta_z)}{f(\xi_\ell)} \right] f(\eta_z) d\eta_z \right\}, \quad (7)$$

where

- $\eta_z := \{\eta_j : z_j = 1\}$ is the vector of probed variables,
- $d_z = \sum_{j=1}^{m} z_j$ is corresponding dimension, i.e., the number of probed variables,
- $f(\xi_\ell)$ is the unconditional likelihood of ξ_ℓ,
- $f(\eta_z)$ is the unconditional likelihood of η_z, and
- $f(\xi_\ell; \eta_z)$ is the conditional likelihood of ξ_ℓ conditional on η_z.

[de: The notation is admittedly rough, but should suffice for now.]
We could re-express (7) as

$$\max_{z\in\{0,1\}^m} \left\{ -\alpha^\top z + q(z) \right\},$$

where $q(z)$ represents a black-box function of z that depends on the offline simulation experiment.

Some remarks about (7):

- The integration in (7) is non-trivial as we likely do not know where the maximum changes. We can circumvent this issue by approximating the integral using Monte Carlo simulation, i.e., generating i.i.d. realizations of η_z, evaluating the integrand, and averaging. The quality of this approximation can be controlled and, importantly, this use of Monte Carlo does not require additional replications of the (expensive) simulator Y.
- Does $q(z)$, or its expected value $q^\circ(z)$ defined later, possess exploitable structure? There is reason to think $q^\circ(z)$ will be monotone increasing and submodular (due to properties of conditional expectation), but $q(z)$ is likely not, because of random sampling.
- The outer optimization in (7) over $z \in \{0,1\}^m$ is also non-trivial, but we could turn to a greedy heuristic or a direct search method to obtain an approximate solution. There should be supporting theory to assess the accuracy of the resulting optimization error/optimality gap.
- There is some reason to believe the likelihood ratio will not explode, which is a common concern and criticism of the likelihood-ratio method. (See imaginary picture of two normal distributions with the probed one having a smaller variance.) Essentially, if the unconditional distribution of ξ has

heavier tails than the conditional distribution of ξ given η_z, e.g., owing to having a larger variance, then the denominator in the likelihood ratio should go to zero slower than the numerator, avoiding issues of dividing by numbers close to zero.

- [de: My plan is to code up a numerical example to see how the solution to (7) compares to that of (1) as n increases. Plus a comparison to the algorithm of [7].]

- The computational advantage we hope to gain by using LR-reweighting is that we do not need to simulate for every z or η_z. This means we can do an offline simulation experiment and then have access to estimates of $q^\circ(z)$ for any and all z. This is different from (or goes beyond) the reasoning for using LR-reweighting in [8].

- We further hope to gain an advantage from using common random numbers (CRN) across alternatives, i.e., using the same $\{\xi_1, \xi_2, \ldots, \xi_n\}$ for the n replications at each x_i, $i = 1, 2, \ldots, n$. Using CRN across alternatives (x_is) is generally advantageous, as it can introduce a positive correlation that reduces the variance in estimators of differences in the performance/objective of different alternatives, which translates to easier optimization. Their use within a LR-reweighting scheme is more complicated, and I don't know of a precedent off the top of my head.

- The LR-reweighting approach should allow us to consider risk measures other than the expectation in the second-stage problem.

- We agreed to look further into settings where different probing decision spaces arise:

  - If m is small, enumeration should be a feasible approach to solving the SAA problem in (7).
  - If m is larger, implicit enumeration methods like branch-and-bound might apply.
  - If the decision space is continuous, who knows?

  We would like to come up with practical examples for these different settings to help focus our efforts on solution approaches.

The decision dependent uncertainty version of (7) is

$$\max_{z\in\{0,1\}^m} \left\{ -\alpha^\top z + \int_{\mathbb{R}^{d_z}} \left[\max_{i=1,2,\ldots,k} \frac{1}{n} \sum_{\ell=1}^{n} Y(x_i, \xi_{i\ell}) \frac{f(\xi_{i\ell}; \eta_z)}{f(\xi_{i\ell})} \right] f(\eta_z) d\eta_z \right\}, \quad (8)$$

where $\xi_{i\ell}$ is a realization of the randomness at alternative x_i, unconditional on η. Compared to (7), the number of simulation replications ($n \times k$) is unchanged, but the number of generated ξs is increased from n to $n \times k$.

The framework in (7) and (8) features equal simulation effort across alternatives; i.e., a common number of replications, n, instead of alternative-dependent sample sizes, n_i. If one regards this n as an initial sample size n_0, then there is an opportunity to allocate additional simulation effort in a way that prioritizes the alternatives (the x_is) that are the most important in determining the anticipated benefit of each probing decision and hence in finding the maximizer of the outer optimization problem over z. For this, we could turn to ideas from R&S/Bayesian optimization/optimal learning to devise acquisition functions. To make this more concrete, we have an inner level problem of estimating

$$q^\circ(z) = \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right]$$

We can reduce the bias of our estimator of $q^\circ(z)$ by increasing the n_i's, but alternatives are not equally important in driving the maximum (and the realizations of η are also weighted differently). Moreover, the (conditional) $Y(x_i, \xi)$ have different variances depending on x_i and η. [de: I think there are a LOT of interesting ideas in this direction (many related to R&S), IF the basic framework shows potential in the numerical experiments.]

[de: Will look more in the literature on sensor location problems, etc. I believe Raghu Bollapragada (or someone else from UT Austin) has done some work in this space.]

5.1 Gradient of the LR estimator

[de: New notes on the gradient.]

Suppose for a moment that the probing variables z are continuous. (We'll put aside the real-world motivations/meanings of continuous probing variables and just do the math.[1]) If we had access to the gradient of the objective function $-\alpha^T z + \cdots$ with respect to z, it could help with the outer optimization over z. We will put aside the simple task of differentiating $-\alpha^T z$ and focus on differentiating

$$q^\circ(z) = \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right].$$

Playing fast and loose with the math,

$$\begin{aligned} \frac{d}{dz} q^\circ(z) &= \frac{d}{dz} \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right] \\ &= \mathbb{E}_\eta \left[\frac{d}{dz} \max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right] \end{aligned}$$

by the Leibniz integral rule. And by the envelope theorem, we can further write

$$\frac{d}{dz} q^\circ(z) = \mathbb{E}_\eta \left[\frac{d}{dz} \mathbb{E}\left[Y(x_{i^*(\eta_z)}, \xi) \mid \{\eta_j \colon z_j = 1\} \right] \right]$$

[1] [bp: Those simulation folks are really shameless.]

where $i^*(\eta_z) = \arg\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i^*, \xi) \mid \{\eta_j : z_j = 1\}\right]$. A variant of the likelihood-ratio method (called the scored-function method) [6] can be used to differentiate the inner expectation, or rather, to construct an unbiased estimator of the derivative of the inner expectation. Skipping several steps, I believe the resulting estimator for the overall derivative will be

$$\widehat{\frac{d}{dz}q^\circ(z)} = \frac{1}{M}\sum_{j=1}^{M}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_{i^*(\eta_z^{(j)})}; \xi_\ell)\frac{f(\xi_\ell; \eta_z^{(j)})}{f(\xi_\ell)}\frac{d}{dz}\log f(\xi_\ell; \eta_z^{(j)})\right], \tag{9}$$

where $\eta_z^{(1)}, \eta_z^{(2)}, \ldots, \eta_z^{(M)}$ are the i.i.d. realizations of all probing variables used to estimate the outer expectation via crude Monte Carlo. We can see from (9) that the gradient estimator comes from "reweighting" the LR estimators that make up our estimate of the inner maximum by their corresponding "scores," shown in pink.

A few notes:

1. (9) should be a biased estimator of $\frac{d}{dz}q^\circ(z)$ due to the non-linearity of the maximum. The bias should vanish as $n \to \infty$.

2. What exactly is the meaning of a probing variable z being continuous? If z is not continuous, can we still use an expression like (9) to get a discrete gradient to help with implicit enumeration? [bp: To be discussed. If $m \leq 10$, probably nothing will beat enumeration. For larger m, purely continuous or mixed integer, I believe the gradient is the only alternative we have!]

3. If applying the chain rule to write

$$\frac{d}{dz}\log f(\xi_\ell; \eta_z^{(j)}) = \frac{d}{d\eta_z^{(j)}}\log f(\xi_\ell; \eta_z^{(j)})\frac{d\eta_z^{(j)}}{dz},$$

   it's not immediately clear how we should define $\frac{d\eta_z^{(j)}}{dz}$. E.g., should we view η_z as a random draw from a distribution parameterized by z and differentiate along the sample path? [bp: ouch! No idea...]

6 Generalization to Continuous Probing and Continuous 2nd-Stage Decisions

Consider a re-write of Equation (1) to accommodate continuous probing (first-stage) variables and continuous decision (second-stage) variables:

$$\max_{z\in\mathbb{R}^m}\left\{-\alpha^T z + \mathbb{E}_\eta\left[\max_{x\in\mathbb{R}^q}\mathbb{E}\left[Y(x,\xi) \mid \eta_z\right]\right]\right\}. \tag{10}$$

A few observations:

- The SAA/LR approach would involve solving the problem

$$\max_{z\in\mathbb{R}^m}\left\{-\alpha^\top z+\int_{\mathbb{R}^{d_z}}\left[\max_{x\in\mathbb{R}^q}\frac{1}{n}\sum_{\ell=1}^{n}Y(x,\xi_\ell)\frac{f(\xi_\ell;\eta_z)}{f(\xi_\ell)}\right]f(\eta_z)d\eta_z\right\}. \tag{11}$$

  Unlike the R&S-type SAA problem, here we cannot run all of the simulation replications offline. To solve the inner max problem, we need to search the continuous x-space, simulating each x when we visit it. This is problematic because when using Monte Carlo integration over η_z, we need to solve the inner problem for each sampled η_z. The issue is further compounded when trying to search over z.

- In other words, generalizing to continuous second-stage variables seems to pose serious tractability problems. In particular, the likelihood-ratio approach is not helping with this issue; LR helps when changing η_z (i.e., the integral) and changing z (i.e., the outer problem).

- As for the outer optimization over z, the LR/score-function gradients could be useful, assuming we can figure out the definition of $d\eta_z/dz$.

7 Basics for a joint model of probes

7.1 Timing and purpose of probing.

The decision problem is two-stage. In the absence of probing, the decision maker must select an alternative (policy) $x \in \{x_1, \ldots, x_k\}$ ex ante, before observing any information beyond the prior model. With probing, the decision maker first selects a subset of probes $z \in \{0,1\}^m$, observes the corresponding signals η_z, and then selects the alternative x as a best response to the observed information. The purpose of probing is precisely to acquire information that can change the identity of the optimal alternative.

7.2 How would a joint model look?

Let ξ denote the primitive randomness driving the discrete-event simulation over the planning horizon $[0,T]$. Fix an "early" observation window $[0,T_0]$ with $T_0 < T$, and decompose

$$\xi = \left(\xi^{(0)}, \xi^{(1)}\right),$$

where $\xi^{(0)}$ generates the sample-path primitives on $[0,T_0]$ and $\xi^{(1)}$ generates the primitives on $(T_0,T]$. [de: $\xi^{(0)}$ can be regarded as the epistemic uncertainty and $\xi^{(1)}$ as the aleatoric uncertainty.] Probes are noisy summaries of the early primitives:

$$\eta_j = g_j\left(\xi^{(0)}\right) + \varepsilon_j, \qquad j = 1, \ldots, m,$$

where ε_j is mean-zero measurement noise and observing η_j costs α_j.

Best response after probing. Given an observed probe vector $\eta_z := \{\eta_j : z_j = 1\}$, the conditional expected performance of alternative x_i is

$$\mu_i(\eta_z) := \mathbb{E}[Y(x_i, \xi) \mid \eta_z],$$

and the second-stage decision is

$$x^\star(\eta_z) \in \arg\max_{i=1,\dots,k} \mu_i(\eta_z).$$

Without probing (i.e., $z = 0$), the decision is instead based on the unconditional expectations

$$x^\star \in \arg\max_{i=1,\dots,k} \mathbb{E}[Y(x_i, \xi)].$$

Probing objective. The first-stage probing decision trades off information value and probe costs:

$$\max_{z\in\{0,1\}^m} \left\{ -\alpha^\top z + \mathbb{E}_{\eta_z}\left[\max_{i=1,\dots,k} \mathbb{E}[Y(x_i, \xi) \mid \eta_z] \right] \right\},$$

optionally subject to a budget constraint $\alpha^\top z \le B$.

7.3 Candidates for function $g(\cdot)$

Examples of early-window probes in a wafer fab.

- **Probe 1: Early demand volume.** Let A_n denote order arrival times and Q_n order sizes. Define

  $$g_1(\xi^{(0)}) = \sum_{n:A_n\le T_0} Q_n,$$

  the total demand volume observed in $[0, T_0]$. The probe reports

  $$\eta_1 = g_1(\xi^{(0)}) + \varepsilon_1.$$

  This represents an early demand assessment based on initial order realizations.

- **Probe 2: Early front-end congestion.** Let $Q^{FE}(t)$ denote the FE queue length process. Define the time-average FE congestion:

  $$g_2(\xi^{(0)}) = \frac{1}{T_0}\int_0^{T_0} Q^{FE}(t)\, dt.$$

  The probe reports

  $$\eta_2 = g_2(\xi^{(0)}) + \varepsilon_2.$$

  This represents a diagnostic report of early front-end congestion.

- **Probe 3: Early back-end utilization.** Let $U^{BE}(t)$ denote the BE utilization rate (fraction of busy servers). Define

$$g_3(\xi^{(0)}) = \frac{1}{T_0} \int_0^{T_0} U^{BE}(t)\, dt.$$

  The probe reports

$$\eta_3 = g_3(\xi^{(0)}) + \varepsilon_3.$$

  This captures early stress in the back-end.

- **Probe 4: Early transfer delay sample.** Let T_k denote realized transfer delays for the first n_0 completed lots. Define

$$g_4(\xi^{(0)}) = \frac{1}{n_0} \sum_{k=1}^{n_0} T_k.$$

  The probe reports

$$\eta_4 = g_4(\xi^{(0)}) + \varepsilon_4.$$

  This represents an early logistics risk indicator.

8 Structural patterns of probing problems

Beyond the wafer-fab setting with a small number of managerial diagnostics, probing models naturally arise in several broader structural patterns. The key distinction is not the application itself, but the geometry of the information acquisition problem.

- **Spatial monitoring (large m or continuous location).**

  In spatial environments, the decision maker chooses where to place sensors or where to allocate monitoring effort over a large geographical region. Examples include wildfire detection, environmental monitoring, and power grid surveillance.

  The probing decision consists of selecting a finite subset of locations or a continuous coverage region under a budget constraint.

  In such settings, the number of potential probe locations can be very large, and continuous placement decisions are natural.

- **Precision or sampling-frequency decisions (continuous probing).**

  In some problems, information quality can be adjusted continuously. The decision variable controls the frequency, resolution, or precision of measurement.

  For instance, in medical monitoring, lab tests or vital measurements can be taken at different frequencies. Higher sampling frequency reduces uncertainty but increases cost.

  Budget constraints can play a role here: there is only a limited number of devices to measure someone's blood pressure continuously.

- **Exploratory investment under uncertainty (discrete but large-scale).**

  In exploration settings such as oil or mineral prospecting, the decision maker chooses which candidate sites to investigate. Each investigation reveals partial information about local geological conditions, which are linked to the eventual payoff of extraction decisions.

  Here the number of candidate sites can be large, and probing corresponds to selective exploratory investment before committing to large-scale operational decisions.

These patterns contrast with the wafer-fab setting considered here, where probes correspond to a small number of early operational diagnostics. In that context, a moderate and finite number of probe options is both realistic and sufficient to generate meaningful value of information.

[de: Some new comments about the nature of probing variables follow.]

In some cases, the epistemic uncertainty $\xi^{(0)}$ can be decomposed into terms that are associated with certain elements of the aleatoric uncertainty. In the wafer-fab setting, some probing variables told us something about demand (order arrivals), while others told us something about service (order fulfillment). I am interested in thinking more about the relationship between probing variables and components of epistemic uncertainty, and the relationship between these components and those of aleatoric uncertainty.

In the case of stochastic simulation, the $\xi^{(0)}$ might naturally represent *parameters* of the distributions that generate the model's random primitives, i.e., the elements that contribute to aleatoric uncertainty, e.g., (inter)arrival times, service times. This pertains the framing of these processes as being "doubly stochastic," i.e., having random parameters.

As for the first question (probing to epistemic), this could be a one-to-one, many-to-one, one-to-many, or many-to-many mapping, depending on the situation. If many-to-one (many probing variables that all tell us something about the common environment), then I suspect some probing variables will clearly dominate others and the problem will not be as interesting.

A Comparison Between Stochastic Programming and Simulation/R&S

This appendix presents a unified notation for stochastic programming and simulation-based ranking and selection. In particular, we define $Y(x, \xi)$ as the realized performance for a decision x under scenario ξ, and $Y(x) = \mathbb{E}[Y(x, \xi)]$ as the expected performance. [de: This is often called $\mu(x)$ in the simulation/R&S area.]

In SP, we take samples that will be *the same for each and every alternative*. Then, as we optimize, we choose the best alternative. Such fact highlights that SP is a *model-based* approach.

Concept	**SP**	**Unified**	**Sim/R&S**
Prim. randomness	ξ	ξ	hidden RNG
One realization	$c^\top x + Q(x,\xi)$	$Y(x,\xi)$	$Y(x)^{(i)}$
Exp. perf.	$c^\top x + \mathbb{E}[Q(x,\xi)]$	$Y(x) = \mathbb{E}[Y(x,\xi)]$	unknown true mean
Sampling	scenarios $\xi^{(i)}$	$Y(x,\xi^{(i)})$	replications $Y(x)^{(i)}$
Optimization	LP/MILP	$\min_x \mathbb{E}[Y(x,\xi)]$	R&S / sequential sampling
Model knowledge	known equations	depends	black-box

B Continuous probing results

Iter	z	$\hat{Q}(z)$	Objective	Gradient
	Starting point: $z^{(0)}$ = [0.5, 0.5, 0.5, 0.5]			
1	[0.5024, 0.5141, 0.4945, 0.4904]	6461.98	6461.98	[47.91, 281.73, -110.07, -191.01]
2	[0.5015, 0.5190, 0.5042, 0.4983]	6462.50	6462.50	[-17.06, 99.00, 194.73, 157.86]
3	[0.5004, 0.5186, 0.5132, 0.5057]	6469.47	6469.47	[-23.74, -8.90, 178.65, 146.60]
4	[0.5011, 0.5230, 0.5212, 0.5025]	6463.97	6463.97	[15.51, 87.63, 160.00, -64.36]
5	[0.5037, 0.5266, 0.5287, 0.5060]	6463.48	6463.48	[51.99, 72.63, 151.25, 70.69]
⋮	⋮	⋮	⋮	⋮
50	[0.5637, 0.7234, 0.7797, 0.5710]	6511.04	6511.04	[147.79, 184.17, -129.19, -36.42]
⋮	⋮	⋮	⋮	⋮
100	[0.6343, 0.8980, 0.9500, 0.6125]	6537.60	6537.60	[27.74, 34.10, 24.23, -11.72]
⋮	⋮	⋮	⋮	⋮
150	[0.6406, 0.9307, 0.9383, 0.6976]	6540.96	6540.96	[110.63, -99.12, -234.94, 132.51]
⋮	⋮	⋮	⋮	⋮
200	[0.6931, 0.9500, 0.9462, 0.7407]	6541.76	6541.76	[-12.21, 182.73, 272.12, 15.16]

Table 1: Optimization trajectory for zero probing cost ($\alpha = 0$) starting from $z^{(0)} = [0.5, 0.5, 0.5, 0.5]$.

Iter	z	$\hat{Q}(z)$	Gradient
0	(0.8000, 0.8000, 0.8000, 0.8000)	–	–
1	(0.7957, 0.8086, 0.8044, 0.8005)	6523.57	(-86.84, 171.05, 88.99, 9.58)
2	(0.7967, 0.8097, 0.8112, 0.7902)	6521.24	(20.37, 23.63, 134.28, -206.10)
5	(0.8061, 0.8165, 0.8206, 0.7980)	6524.29	(-39.81, 102.26, 135.36, -30.02)
10	(0.8019, 0.8370, 0.8290, 0.8090)	6520.56	(-87.78, -31.99, 71.91, 3.29)
20	(0.7961, 0.8812, 0.8476, 0.8237)	6536.02	(10.22, -91.59, 76.00, 4.86)
40	(0.8512, 0.9500, 0.9127, 0.8294)	6549.61	(-35.95, 151.74, 172.24, 60.65)
60	(0.8528, 0.9335, 0.9500, 0.8298)	6544.07	(-110.83, -57.23, 271.70, -62.94)
80	(0.8934, 0.9182, 0.9296, 0.8725)	6544.62	(37.11, -127.11, -210.11, -81.94)
100	(0.9286, 0.9494, 0.9500, 0.9168)	6557.98	(-16.97, 112.35, 372.12, 140.17)
120	(0.9177, 0.9500, 0.9500, 0.9211)	6554.19	(152.78, 229.22, 123.90, -48.48)
140	(0.9500, 0.9385, 0.9500, 0.9357)	6545.60	(27.32, 78.03, 160.56, -174.91)
160	(0.9243, 0.9500, 0.9461, 0.9308)	6550.14	(-104.11, 13.16, -77.13, 52.03)
180	(0.9477, 0.9452, 0.9500, 0.9456)	6559.36	(27.41, 147.92, 80.84, 17.06)
200	(0.9325, 0.9500, 0.9500, 0.9196)	6549.65	(121.88, 30.40, 29.98, -57.61)

Table 2: Stochastic gradient optimization starting from $z^0 = (0.8, 0.8, 0.8, 0.8)$ and no probing costs.

Iter	z	$\hat{Q}(z)$	Objective	Gradient
1	[0.7942, 0.8066, 0.8019, 0.7975]	6523.57	6379.57	[-116.84, 131.05, 38.99, -50.42]
2	[0.7927, 0.8059, 0.8079, 0.7845]	6522.60	6378.56	[-28.16, -13.40, 119.05, -258.59]
3	[0.8024, 0.8061, 0.8043, 0.7954]	6531.76	6388.28	[192.74, 4.76, -72.44, 216.64]
4	[0.8016, 0.8021, 0.8079, 0.7893]	6516.19	6371.93	[-15.82, -79.46, 72.74, -121.43]
5	[0.7968, 0.8002, 0.8140, 0.7938]	6525.09	6381.20	[-96.36, -38.08, 120.78, 89.24]
6	[0.7999, 0.8132, 0.8237, 0.7917]	6531.65	6387.41	[63.32, 259.35, 193.92, -42.06]
7	[0.7909, 0.8182, 0.8267, 0.7858]	6523.94	6378.73	[-179.89, 100.58, 61.27, -117.35]
8	[0.7907, 0.8242, 0.8229, 0.7888]	6532.23	6387.29	[-4.68, 119.85, -76.74, 60.90]
9	[0.7895, 0.8237, 0.8140, 0.7899]	6522.91	6377.74	[-24.45, -9.70, -176.60, 21.94]
10	[0.7880, 0.8268, 0.8113, 0.7922]	6520.18	6375.45	[-30.80, 61.28, -54.81, 45.55]
⋮	⋮	⋮	⋮	⋮
200	[0.7558, 0.9413, 0.9492, 0.2160]	6531.09	6410.10	[14.80, -76.32, 311.74, -298.65]

Table 3: Gradient-based optimization of $Q(z) = \mathbb{E}[\max_i Y(x_i, \eta; z)]$ using likelihood ratio estimation. $\alpha = (30.0, 40.0, 50.0, 60.0)$ and $z^0 = (0.8, 0.8, 0.8, 0.8)$

Final solution:

$$z^* = [0.7558,\ 0.9413,\ 0.9492,\ 0.2160].$$

C Stochastic gradient computation

Objective and estimator. The objective is

$$Q(z) = \mathbb{E}\big[\max_i Y(x_i, \eta_z)\big],$$

where η_z denotes all randomness whose distribution depends on z. For each replication r, the simulator generates one sample path $\eta^{(r)}$ and evaluates all

policies on that same path:

$$Y_r^{\max} = \max_i Y(x_i, \eta^{(r)}).$$

The goal is to estimate $\nabla Q(z)$.

Likelihood-ratio identity. Using the score-function method,

$$\nabla Q(z) = \mathbb{E}[Y^{\max}(\eta_z)\, \nabla_z \log f_z(\eta_z)]\,,$$

where f_z is the joint density of all z-dependent random events along the sample path.

Factorization over jobs. Only three mechanisms depend on z:

1. probing decision for class c: $p_{\text{probe}}(z, c) = z_c$;
2. signal generation (if probed), with probabilities $p_h(c, z)$ or $p_f(c, z)$;
3. disruption in standard BE (for complex jobs): $p_d(c, z) = 1 - 0.75 z_c$.

Hence the log-likelihood decomposes additively over jobs:

$$\nabla_z \log f_z(\eta) = \sum_n \nabla_z \log P_z(\text{events for job } n).$$

Bernoulli score. For $B \sim \text{Bernoulli}(p(z))$ and observation $b \in \{0, 1\}$,

$$\nabla_z \log P(B = b) = \left(\frac{b}{p(z)} - \frac{1 - b}{1 - p(z)} \right) \nabla_z p(z).$$

Model-specific derivatives. The required derivatives are:

$$\frac{\partial p_{\text{probe}}}{\partial z_c} = 1, \qquad \frac{\partial p_d}{\partial z_c} = -0.75,$$

$$\frac{\partial p_h}{\partial z_c} = (1 - p_c^{\text{hit}}) \beta e^{-\beta z_c}, \qquad \frac{\partial p_f}{\partial z_c} = -p_c^{\text{false}} \beta e^{-\beta z_c}.$$

Each observed event contributes a term of the Bernoulli form multiplied by these derivatives.

Replication-level gradient. For replication r, let

$$S_r(z) = \nabla_z \log f_z(\eta^{(r)})$$

be the total score (sum over all jobs). The likelihood-ratio estimator is

$$G_r(z) = Y_r^{\max}\, S_r(z).$$

Variance reduction and final estimator. Using a baseline $\bar{Y} = \frac{1}{R}\sum_r Y_r^{\max}$, the estimator becomes

$$\widehat{\nabla Q(z)} = \frac{1}{R}\sum_{r=1}^{R}(Y_r^{\max} - \bar{Y})\, S_r(z).$$

With a linear cost term $-\alpha^\top z$, the final gradient estimate is

$$\widehat{\nabla(Q(z) - \alpha^\top z)} = \frac{1}{R}\sum_{r=1}^{R}(Y_r^{\max} - \bar{Y})\, S_r(z) - \alpha.$$

Algorithmic use. At each iteration, independent sample paths are generated, the best policy is selected *within each path*, and the gradient is estimated using the corresponding $Y_r^{\max}$. The resulting estimator is used in a projected stochastic-gradient update for z.

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